Mateus Piovezan Otto

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Education

University of Washington Seattle, WA
Ph.D. in Statistics Fall 2024 -

Coursework: Statistical Theory, Regression Models, Design of Medical Studies, Survival Analysis.

University of São Paulo & Federal University of São Carlos São Carlos, BR
M.Sc. in Statistics 2021-2023

Coursework: Measure Theory, Statistical Machine Learning, Stochastic Processes.

University of São Paulo São Carlos, BR
B.Sc. in Computational Physics 2017-2021

Coursework: C++ Programming, High Performance Computation, Statistical Physics.

Research Experience

University of Washington Seattle, WA
Research Assistant, Supervisor: Prof. Zaid Harchaoui Winter 2025 - Spring 2026

Zero-shot prediction with multimodal contrastive learning objectives. Incremental and supervised methods for Independent Component Analysis (ICA). Randomized algorithms for approximate diagonalization and applications to ICA.

University of São Paulo & Federal University of São Carlos São Carlos, BR
Research Assistant, Supervisor: Prof. Rafael Izbicki 2022-2023

Regression trees for prediction intervals with conditional coverage guarantees [1].

University of São Paulo & Federal University of São Carlos São Carlos, BR
Research Assistant, Supervisor: Prof. Rafael Izbicki 2021-2023

Scalable and interpretable kernel methods with random Fourier features [0, 3].

University of São Paulo São Carlos, BR
Research Assistant, Supervisor: Prof. Aion da Escóssia Melo Viana 2020-2021

Dark matter phenomenology and search in the Galactic Center with the Fermi LAT [4].

Publications

Journals

[1] Luben M.C. Cabezas, Mateus P. Otto, Rafael Izbicki, and Rafael B. Stern. "Regression trees for fast and adaptive prediction intervals". In: *Information Sciences* 686 (2025), p. 121369.

[2] Mateus Piovezan Otto and Rafael Izbicki. "RFFNet: large-scale interpretable kernel methods via random Fourier features". In: *Computational Statistics* 41.7 (Nov. 2026).

Theses

[3] Mateus P. Otto. "Scalable and interpretable kernel methods based on random Fourier features". Master's thesis. USP, 2022.

[4] Mateus P. Otto and Aion da Escóssia Melo Viana. "Searching for dark matter in the Galactic Center with the Fermi Gamma-ray Space Telescope". Senior thesis. IFSC, 2020.

Teaching Experience

University of Washington

Teaching Assistant, Graduate Courses

Advanced Methods of Statistical Inference I (STAT 581/BIOST 583)	Fall 2026
Statistical Machine Learning for Data Scientists (DATA/STAT/BIOST 558)	Spring 2026
Introduction to Statistics and Probability (DATA/STAT/BIOST 556)	Fall 2024

Talks

A randomized Jacobi-type algorithm for approximate diagonalization — January 2026
Institute for Foundations of Data Science, Seattle, WA

Scalable and interpretable kernel methods based on random Fourier features — November 2022
Statistical Machine Learning Lab, São Carlos, BR

Scalable and interpretable kernel methods based on random Fourier features — August 2022
Group of Inductive Statistics (GIS), São Carlos, BR

Fellowships and Awards

Fellowships

FAPESP Master's Fellowship	2021-2023
CNPq Undergraduate Research Fellowship	2020-2021
FAPESP Undergraduate Research Fellowship	2019-2020

Awards

Best Poster Award, Pacific Northwest SIAM Conference	2025
Bernhard Gross Prize, University of São Paulo (≡ *Summa cum laude*)	2020
Bernhard Gross Prize, University of São Paulo (≡ Student of the Year)	2018, 2019
Gold Medal, Brazilian Astronomy Olympiad	2015

Summer Schools

Computation, Logic, and Set Theory — Rio de Janeiro, BR
Instituto de Matemática Pura e Aplicada (IMPA), January–March 2024

Metric Spaces — Online
Instituto de Matemática e Estatística, January–February 2021

Probability Theory — São Carlos, BR
Instituto de Ciências Matemáticas e de Computação, January–February 2019